

Likelihood-Ranked Synthetic Selection in Self-Consuming Diffusion: Minority Extinction Ordering on Imbalanced Mixtures

Muhammad Alamzeb
Independent researcher
shayankhanmahar@gmail.com

Abstract

Self-consuming training loops—retraining generative models on their own outputs—can induce *model collapse*, including loss of rare modes. We study unlabeled selection of synthetic samples by a denoising likelihood proxy in self-consuming DDPM training on imbalanced mixtures. Under matched real-mix ratio and matched synthetic budget, the first-generation minority-mass ratio $r = m^{(1)}/m^{(0)}$ obeys

$$\text{bottom-}k > \text{random-}k > \text{top-}k$$

on a primary matrix of **eight** seed–imbalance cells (seeds $\{0, 1, 2\}$ at $\rho \in \{5, 10\}$ plus seeds $\{3, 4\}$ at $\rho=5$ only; means 0.57/0.36/0.17). Paired contrasts vs. random- k show large effects (Cohen’s $d_z \approx 1.0$) with Wilcoxon $p \approx 0.042$ (Holm-adjusted $p \approx 0.085$ for the two tests); we treat the ordering as supported by effect size and directional consistency, with small- n caveats. Likelihood top- k also concentrates majority mass relative to random- k . A hypothesized “false stability” pattern—top- k *improving* sliced W_2 while minorities die—is **not observed**: at generation 1, $\Delta W_2(\text{top} - \text{rand}) > 0$ on 8/8 cells (Wilcoxon $p \approx 0.014$). Directional transfer checks on 8D GMM and Digits-PCA (3 seeds each) match the same order. We release configs, seeds, logs, and analysis code with the manuscript package.

1 Introduction

As synthetic data proliferates, generative models are increasingly trained on mixtures of real and model-generated samples. Iterated self-consumption can cause *model collapse*: progressive loss of diversity and forgetting of rare modes [Shumailov et al., 2023, Alemohammad et al., 2023, Gerstgrasser et al., 2024]. Mitigations include accumulating real data [Gerstgrasser et al., 2024], injecting fresh real samples, and *filtering* synthetic data [Feng et al., 2024, Cai et al., 2025].

When the only available score is the generator’s own likelihood (or denoising loss), how should one select synthetics? Feng et al. [2024] show that *task verifiers* for labeled synthetic answers can prevent collapse. Cai et al. [2025] filter “unrealistic” diffusion samples via latent probes and report aggregate FID/precision/recall. Alemohammad et al. [2023] observe that sampling bias toward high-quality generations drives a quality–diversity tradeoff. What is missing is a **matched-budget**, **mode-exact** causal contrast among likelihood ranks.

What is new. Prior work describes quality–diversity tradeoffs and filtering heuristics. We isolate an *interventional* selection policy π (top- k vs. random- k vs. bottom- k) at matched synthetic count and matched real-mix α , and measure mode masses against known centers. The contribution is this controlled contrast and the resulting empirical minority-retention order—not a new named training algorithm or ImageNet-scale SOTA method.

Contributions.

1. A controlled self-consuming DDPM protocol on imbalanced GMMs with exact mode masses.
2. Evidence for a likelihood-rank ordering of minority retention under matched k and α .
3. Evidence that top- k does *not* improve sliced W_2 vs. random- k while minorities decline (the opposite of a “better aggregate, worse minority” dissociation).
4. Directional transfer checks: 8D GMM and Digits-PCA with nearest class-mean occupancy.

2 Related work

Model collapse. Shumailov et al. [2023] document recursive training collapse and tail forgetting. Gerstgrasser et al. [2024] show accumulating real+synthetic data can avoid collapse. Alemohammad et al. [2023] (*MAD*) highlight quality-biased sampling.

Verification and filtering. Feng et al. [2024] argue verification of synthesized *labels* prevents collapse. Cai et al. [2025] (*LSF*) filter synthetic images using latent-space realism scores. Neither isolates unlabeled ELBO top- k vs. random- k with mode-grounded metrics on continuous mixtures.

Temperature / diversity knobs. Inference-time temperature methods [Xu et al., 2025, Li et al., 2026] reshape sampling diversity; we instead intervene on *training-set selection* inside self-consuming loops.

3 Method

3.1 Self-consuming loop

Let $f^{(g)}$ be a DDPM trained on $\mathcal{D}^{(g)}$. Sample synthetic set $\widehat{\mathcal{D}}^{(g+1)}$. Build the next train set by mixing a fixed real buffer $\mathcal{D}_{\text{real}}$ (fraction α) with selected synthetics.

3.2 Likelihood proxy and policies

Score $s(x)$ is mean denoising MSE over a t -grid (lower = higher likelihood). At matched synthetic count n_{syn} and real fraction α :

- **top- k / bottom- k / random- k :** select a pool of size $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}} N \rfloor)$ by lowest / highest / uniform scores, then draw n_{syn} from that pool and mix with α real.
- **mix:** same α real mix, but the synthetic portion is a uniform subsample of *all* N synthetics (no rank filter; no k -pool stage).
- **replace:** training set is pure synthetic (collapse reference).

Thus **mix** $\neq$ **random- k** : random- k still uses a size- k subsample stage (matched to top/bottom); mix skips ranking entirely.

Table 1: Primary analysis cells (exactly eight seed–imbalance pairs). Seeds 3–4 were run only at $\rho=5$ (replication); $\rho=10$ uses seeds $\{0, 1, 2\}$ only.

$\rho \backslash$ seed	0	1	2	3	4
5	✓	✓	✓	✓	✓
10	✓	✓	✓	—	—

Table 2: Minority-mass ratio $r = m^{(1)}/m^{(0)}$ on the primary eight-cell matrix (Table 1).

Policy	Mean r	Std	n
bottom_k	0.572	0.289	8
rand_k	0.360	0.173	8
mix	0.328	0.100	8
top_k	0.172	0.078	8

3.3 Metrics

Nearest-mean mode assignment on known centers (GMM means, or Digits empirical class means in PCA space). Report minority-mean mass m (modes $1 \dots K - 1$), majority mass (mode 0), and the ratio

$$r = \frac{m^{(1)}}{m^{(0)}},$$

which we call *retention* even when $r > 1$ (minority mass can rise after one generation under bottom/random selection). Sliced W_2 : average of squared 1D Wasserstein distances over **24** random projections; each comparison uses up to **400** synthetic and **400** held-out real points (same seed schedule as the run).

4 Experiments

4.1 Primary design (defines $n=8$)

$K=4$ modes, tiny MLP denoiser, $G=5$, $\alpha=0.5$, $k_{\text{frac}}=0.5$. Logs: `w2_fs.jsonl` ($n=228$ records).

4.2 Primary retention results

Table 2 averages $r = m^{(1)}/m^{(0)}$ over the eight cells in Table 1.

At $\rho=5$, generation 1, all five seeds satisfy bottom $>$ rand $>$ top on minority mass. Paired retention contrasts vs. random- k ($n=8$): top–rand mean $\Delta=-0.188$ (bootstrap 95% CI $[-0.30, -0.06]$; Cohen’s $d_z=-1.04$; Wilcoxon $p \approx 0.042$); bottom–rand mean $\Delta=+0.212$ (CI $[0.08, 0.35]$; $d_z=+1.00$; $p \approx 0.042$). With two tests against the same baseline we apply Holm correction: both adjusted $p \approx 0.085$ (above 0.05). We therefore emphasize **effect sizes**, CIs, and directional counts (7/8 top worse than rand; 6/8 bottom better) rather than uncorrected p -values alone. Top- k also raises majority mass vs. random- k at early generations.

4.3 Sliced W_2 (“false stability” check)

If top- k were “falsely stable,” we would expect better (lower) sliced W_2 than random- k while minorities die. Instead, at generation 1, $\Delta W_2(\text{top} - \text{rand})$ has mean $+0.113$ (std 0.099), positive on

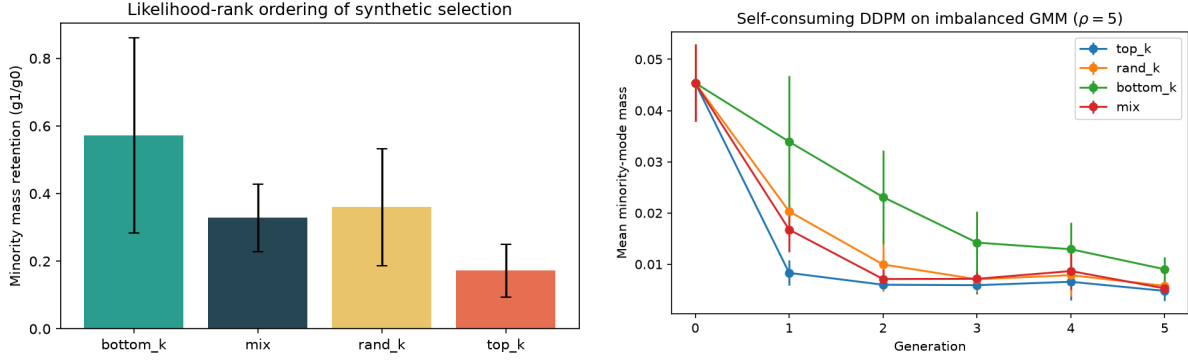


Figure 1: Left: retention by policy on the primary matrix. Right: minority mass vs. generation at $\rho=5$.

Table 3: Mean retention r by real-mix α (two seeds). Order = bottom>rand>top on both seeds.

α	bottom	rand	top	Order holds
0.25	0.25	0.17	0.05	1/2
0.50	0.62	0.49	0.24	2/2
0.75	1.34	0.77	0.94	0/2

all eight cells (Wilcoxon $p \approx 0.014$). Means remain positive for generations 2–5 (6–7 of 8 cells; see `table_w2_top_minus_rand.csv`). We report this as **evidence against** a top- k W_2 improvement under minority loss—not as a formal Neyman–Pearson “falsification” of a fully specified alternative.

4.4 Real-mix sensitivity (α)

Separate ablation (`w2_fs_alpha.jsonl`; seeds $\{0, 1\}$, $\rho=5$, $G \leq 3$):

At $\alpha \leq 0.5$, top- k remains worst; at $\alpha=0.75$ the top vs. random gap reverses on average while bottom stays highest—an important scope condition ($r > 1$ again means minority mass rose vs. $g=0$).

4.5 Transfer checks (directional; $n=3$)

These are **not** powered significance tests; we report mean $\pm$ std and seed-wise order counts.

- **8D GMM** (`w2_fs_hd`): retention 0.55 ± 0.20 / 0.27 ± 0.06 / 0.08 ± 0.03 (bottom/rand/top); order holds 3/3.
- **Digits-PCA** (`w2_fs_digits_pca`): 1.49 ± 0.11 / 1.11 ± 0.08 / 0.68 ± 0.17 ; order holds 3/3. Ratios > 1 mean minority mass increased from $g=0$ to $g=1$ under bottom/random (definition of r above).

An earlier Digits CVAE+classifier probe is inconclusive and unused for claims.

4.6 Selection-pool size (k_{frac})

Under $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}} N \rfloor)$, $k_{\text{frac}} \in \{0.25, 0.5\}$ are operationally equivalent when n_{syn} binds; at 0.75 top- k remains worst.

5 Analysis

Likelihood top- k preferentially retains points the model already explains well (majority modes), reducing minority inflow. Bottom- k does the opposite relative to random. Aggregate sliced W_2 is a poor monitor for minority survival: it moves with top- k in the wrong direction for a “quality win.”

6 Limitations

- Toy 2D/8D GMMs and Digits-PCA with tiny networks; not ImageNet-scale.
- Primary Wilcoxon tests use only $n=8$ pairs; uncorrected $p \approx 0.042$ becomes Holm-adjusted $p \approx 0.085$. Inference relies on large d_z , CIs, and directional consistency.
- Transfer checks use three seeds without formal hypothesis tests.
- Digits CVAE transfer was inconclusive; Digits-PCA uses embeddings, not raw pixels.
- Effect strongest at moderate imbalance and early generations; high real mix ($\alpha=0.75$) weakens top vs. random.
- Contribution is a controlled empirical contrast relative to MAD/LSF/Feng, not a new SOTA training method.

7 Conclusion

Unlabeled likelihood ranking of synthetic data in self-consuming diffusion induces a reproducible minority-retention *order* (bottom- $k > \text{random-}k > \text{top-}k$) under matched budgets on imbalanced mixtures, with large paired effect sizes despite small n . Top- k does not improve sliced W_2 relative to random- k in our primary matrix. Preferring “most likely” synthetics is not a free lunch for minority survival.

References

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A Reproducibility and code

Cross-platform entry point: `python scripts/reproduce_w2.py` (also `scripts/reproduce_w2.sh` and Windows `scripts/reproduce_w2.ps1`). All numerical tables are produced from logged JSONL via analysis scripts (no hand-edited metrics). Experiment IDs: `w2_fs`, `w2_fs_hd`, `w2_fs_digits_pca` (supported); `w2_fs_digits` (CVAE; inconclusive). Source, configs, seeds, and logs are included in the project archive accompanying this manuscript (commit hash recorded in `SUBMIT.md`); a public GitHub/Zenodo URL will be linked upon posting.